

A

Internship Report

On

“Web Development”

Submitted to the
Department of Information Technology

By

Akshit Kotnala

Class: TE IT

Roll No: 4331

Exam No: T400220301

Under the supervision of
Internal Guide: Prof. Sandeep Samleti



Department Of Information Technology

Army Institute of Technology, Pune

2024-2025

Semester: II



CERTIFICATE

This is to certify that the internship report entitled **“Web Development”** being submitted by **Akshit Kotnala (4302)** is a record of Bonafide work carried out by her with prior permission of HOD IT under the supervision of **Prof. Sandeep Samleti** in the academic year 2024-2025.

Date: 14/05/2025

Place: Pune

Prof. Sandeep Samleti
Internship Guide

Dr. Sangeeta Jadhav
Head of the Department

CERTIFICATE BY THE COMPANY/OFFER LETTER



INTERNSHIP OFFER LETTER

Date : 23/01/2025

ID:CS25NY378561

Dear,

Akshit Kotnala

We would like to congratulate you on being selected for the "Web Development" virtual internship position with "CodSoft". We at CodSoft are excited that you will join our team.

The duration of the internship will be of **4 weeks**, starting from **25 January 2025 to 25 February 2025**. The internship is an educational opportunity for you hence the primary focus is on learning and developing new skills and gaining hands-on knowledge. We believe that you will perform all your tasks/projects.

As an intern, we expect you to perform all assigned tasks to the best of your ability and follow any lawful and reasonable instructions provided to you.

We are confident that this internship will be a valuable experience for you, we look forward to working with you and helping you achieve your career goals.

By accepting this offer, you commit to executing assigned tasks diligently and ensuring excellence in all aspects of your work.

Best of Luck!

Thank You!

Founder (CodSoft)



MSME Registered

www.codsoft.in
contact@codsoft.in



ACKNOWLEDGEMENT

I would like to express my heartfelt gratitude to everyone who supported and guided me during the successful completion of my internship at **CodSoft**. This journey has significantly contributed to both my technical growth and personal development.

First and foremost, I would like to thank **Prof.Sandeep Samleti**, my Internal Guide, for their constant encouragement, valuable insights, and continuous support throughout this internship. Their guidance helped me stay focused, overcome challenges, and complete the assigned projects effectively.

I am sincerely thankful to the **CodSoft team** for offering me this opportunity to explore the world of web development in a practical and engaging environment. Working independently on real-world projects like landing pages, a personal portfolio, and a book purchasing website helped me strengthen my understanding of frontend and backend development using technologies like **HTML, CSS, JavaScript, React, and MongoDB**.

Lastly, I extend my gratitude to all those who directly or indirectly contributed to the success of this internship. This experience has not only enhanced my technical skills but also instilled a sense of responsibility, creativity, and confidence in my professional capabilities.

Name: Akshit
Roll No: 4302
Exam No: T400220301

Abstract

This report provides a concise overview of my internship experience at **CodSoft**, where I undertook multiple web development projects independently. The primary objective of the internship was to gain hands-on experience in full-stack web development and strengthen my practical understanding of frontend and backend technologies.

Throughout the internship, I developed several projects including a **responsive landing page**, a **personal portfolio website**, a **functional calculator app**, and a **book purchasing platform**. These were built using **HTML, CSS, and JavaScript**, with advanced implementation involving **React.js** for interactive UI and **MongoDB** for backend database management. Animations and responsive design were key focus areas to ensure a modern user experience.

Each project followed a modular development process — starting from requirement gathering and layout structuring to component styling, functionality integration, debugging, and documentation. Version control was managed using **Git and GitHub**, enabling organized and traceable progress. I also explored **UI/UX principles**, animations, and mobile responsiveness throughout the development phase.

Working independently allowed me to take full ownership of every task and enhanced my problem-solving, time management, and self-discipline. This report outlines my weekly tasks, tools and technologies used, key takeaways, and the impact this internship has had on my technical and professional development.

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Chapter No 1

INTRODUCTION

The web development internship at CodSoft offered an enriching opportunity to convert academic learning into real-world application. As a third-year Information Technology student, I gained practical exposure to modern web technologies, UI/UX design strategies, and complete software project workflows. The internship allowed me to independently build multiple web-based projects from scratch, enhancing both technical proficiency and confidence.

During the course of the internship, I successfully completed two major tasks:

- **A Drum Website** that allows users to play virtual drums by clicking or tapping on drum pads, offering an engaging and interactive musical experience.
- **An E-commerce Clone** that replicates core features of platforms like Amazon, including product listings, shopping cart, and user interface, built using modern web technologies.

These projects gave me exposure to both front-end development using web standards and full-stack integration with database systems. I also enhanced my design sense by incorporating responsive layouts, animations, and smooth user interaction.

1.1 Learning Objectives of Internship

The primary goal of this internship was to bridge theoretical knowledge with industrial web development practices and to enhance creativity and technical fluency through project-based learning. The main learning objectives included:

- **Apply classroom knowledge:** Reinforce and apply concepts of HTML, CSS, JavaScript, React, and MongoDB.
- **Build practical projects:** Independently develop complete websites with frontend and backend features.
- **Explore UI/UX Principles:** Improve understanding of layout, user flow, responsiveness, and interactive design.
- **Gain hands-on with tools:** Work with tools like VS Code, GitHub, Chrome DevTools, and Figma.
- **Work in a time-bound environment:** Improve productivity, deadline management, and project documentation.
- **Boost professional readiness:** Build portfolio-quality projects and gain confidence for industry-level work.

Each objective was aligned with my academic and career aspirations, and the learning environment at CodSoft provided the resources and guidance needed to achieve them.

1.2 Module Description

The internship was divided into multiple modules that spanned the 6-week duration, offering structured learning around both frontend and full-stack development.

Module 3: Drum Website (Interactive Sound Application)

This module focused on building an interactive website that plays drum sounds when users click or press specific keys. It was designed for an engaging user experience with instant audio-visual feedback.

Key learnings included:

- Event handling in JavaScript for click and keypress actions.
- Integrating sound files to play tunes on interaction.
- Animating drum pads for visual feedback using CSS.
- Creating a responsive layout for mobile and desktop compatibility.
- Enhancing user interaction with transitions and delay effects.

This project helped strengthen my JavaScript fundamentals and introduced me to interactive web audio development.

Module 4: E-commerce Clone (Amazon-like Website)

This module involved creating a mini version of a popular e-commerce platform, mimicking features like product display, cart, and user interface.

Key learnings included:

- Creating dynamic product listings with JavaScript and React.
- Implementing add-to-cart and item count functionality.
- Designing a responsive, user-friendly UI using CSS and Flexbox.
- Simulating checkout and user login flows.
- Managing component state and props in React for better structure.

This project gave me a strong understanding of front-end e-commerce workflows and improved my ability to replicate real-world applications.

Chapter No 2

TECHNOLOGY

The web development internship at **CodSoft** gave me the opportunity to work with a variety of technologies and tools commonly used in the software industry. These tools enabled me to design, develop, test, and deploy responsive and interactive web applications. Each project undertaken during the internship was built using a curated set of technologies selected based on functionality, scalability, and user experience needs.

The following is an overview of the technologies and tools used:

Frontend Technologies:

Technology	Purpose / Use
HTML5	Used for structuring content on the web pages using semantic elements.
CSS3	Applied for styling, layout, and animations to make the website visually appealing.
JavaScript	Added dynamic behavior to websites such as form validation and interactive elements.
React.js	Used in the Book Purchasing Website for creating reusable UI components and managing frontend logic.
Bootstrap	For faster UI development and responsive layout structures.

Backend / Database:

Technology	Purpose / Use
MongoDB	NoSQL database used to store book information and user data in the Book Purchasing Website.

Design & Development Tools:

Tool	Purpose / Use
Visual Studio Code (VS Code)	Main IDE used for writing, testing, and debugging code.
Git & GitHub	Version control system used to manage codebase, track changes, and collaborate efficiently.
Figma	Used for designing initial layouts and UI components for structured prototyping (if required).
Chrome Developer Tools	Assisted in debugging and optimizing frontend performance across browsers.

Other Technologies & Concepts:

Concept / Tool	Purpose / Use
Media Queries	Enabled responsive web design for different screen sizes and devices.
CSS Animations & Transitions	Used in the portfolio site to add visual effects and improve interactivity.
REST APIs	For data exchange between frontend and backend (basic simulation in the Book Purchasing Project).

Chapter No 3

SCREENSHOTS OF RESULTS

05



Search...





Your Cart

Acer SB220Q bi 21.5 inches Full HD (1920 x 1080)
IPS Ultra-Thin



\$599.0

+ 1 -

 Save  Delete

Rain Jacket Women Windbreaker Striped
Climbing Raincoats



\$40.0

Size: S

+ 1 -

 Save  Delete

Congrats! You're eligible for **Free Delivery**.
Use code **SHUBHO20** for 20% discount.

Promocode

Apply

Sub-Total **\$638.99**

Delivery \$0.00

Tax (5%) + \$31.95

Total **\$670.94**

Proceed to Payment



Sign in

User Registration

Hey, Enter your details to create a new account

Please fill in this field.

Sign up

— Or —

Sign up with 





Search...





Fjallraven - Foldsack No. 1 Backpack..
MEN'S CLOTHING
★★★★★ 5 (110 reviews)
\$109.95 ~~\$183~~ (60% OFF)



Mens Casual Premium Slim Fit T-S..
MEN'S CLOTHING
★★★★★ 5 (83 reviews)
\$22.3 ~~\$37~~ (60% OFF)



Mens Cotton Jacket
MEN'S CLOTHING
★★★★★ 5 (122 reviews)
\$55.99 ~~\$93~~ (60% OFF)



Mens Casual Slim Fit
MEN'S CLOTHING
★★★★★ 5 (145 reviews)
\$15.99 ~~\$27~~ (60% OFF)



John Hardy Women's Legends Naga ..
JEWELERY
★★★★★ 5 (124 reviews)
\$695 ~~\$1154~~ (60% OFF)











Chapter No 4

CONCLUSION

The internship at **CodSoft** has been an enriching and insightful experience, enabling me to apply theoretical concepts in practical scenarios. As a third-year Information Technology student, this opportunity allowed me to strengthen my web development skills, gain industry-level exposure, and grow both technically and professionally.

During the course of this internship, I successfully completed two additional projects that further enhanced my skills in interactive web development and frontend e-commerce design:

1. **Drum Website (HTML, CSS, JavaScript):** I developed an interactive drum kit web application where users can play drum sounds by clicking buttons or pressing keys. This project improved my understanding of JavaScript event handling, audio integration, and responsive design. It also strengthened my skills in creating engaging, interactive UI experiences with visual and sound feedback.
2. **E-commerce Clone (HTML, CSS, JavaScript, React):** I built a simplified e-commerce clone inspired by platforms like Amazon. The project included product listings, a shopping cart, and responsive layouts. I used React for component-based development and JavaScript for cart logic, simulating real-world e-commerce functionalities. This helped me gain hands-on experience with modern frontend frameworks and user-focused design.
3. **Weather Forecast Website (HTML, CSS, JavaScript, API Integration):** I developed a weather forecast website that displays real-time weather information for any city entered by the user. It involved integrating a third-party weather API to fetch live data.

Throughout the internship, I became more comfortable with **HTML5**, **CSS3**, and **JavaScript**, and gained a better understanding of **responsive web design**, **form validation**, and **client-side data management**. Additionally, I improved in **debugging techniques**, **version control using Git**, and maintaining code quality.

Beyond technical learning, the internship also helped me enhance my **time management**, **problem-solving**, and **self-discipline**. Working independently on real-world tasks boosted my confidence and prepared me for future challenges in the tech industry.

I am thankful to **CodSoft** for providing a structured and motivating environment where I could grow as a developer. The experience gained from these projects has laid a solid foundation for my career in web development and beyond.

Chapter No 5

COMPLETION CERTIFICATE



Chapter No 6

Internship Feedback

6.1 Student Internship Feedback

**Army Institute of Technology
Department of Information Technology
AY 2024-25 Sem: I / II**

Name of the Student: Akshit Kotnala

Class: TE IT

Name of the Company/Organization where you have done the Internship: CodSoft

Note: Provide following feedback by answering the following questions either Yes or No.

Q.1 Is this internship increases your technical skill? (Yes/No):

Ans: Yes

Q.2 Is the mentor from the company helped you in the Project? (Yes/No)

Ans: Yes

Q.3.Do you fill that this internship will be helpful for you in future? (Yes/No)

Ans: Yes

Q.4 Is our college has provided the enough support to do this Internship? (Yes/No)

Ans: Yes

Any suggestions:

No Suggestions.

**Akshit
Roll No: 4302
Exam No: T400220301**

Chapter No 7

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